



INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR

Time Card for Social Service or Community Work

Name of Student: Masalge Harshvardhan Umesh Roll. No. :24110196

Details of Activity/ Voluntary service: Library

Name of Supervisor: Sheetal Tank

Sr. No.	Date	In-Time	Out-Time	No. of Hours	Description of work carried out with Remarks (if any)
1	04/05/2026	10:00 PM	2:00 AM	4	Background research: Searched across various library websites and studied how their chatbots work. Proposed three different architectures of chatbots for this particular task based on
	05/05/2026	5:00 PM	6:30 PM	1.5	Preprocessing and exploring the dataset
	05/05/2026	11:00 PM	3:00 AM	4	Developed the Data Ingestion & Indexing Pipeline: Created an offline, automated data pipeline using Pandas to ingest the raw Excel journal database. Developed logic to normalize text, concatenate metadata fields (titles, subjects, publishers) into comprehensive embedding strings, and systematically populate a local SQLite database with structured records.
	06/05/2026	12:00 PM 4:00 PM	2:00 PM 7:00 PM	5	Vector Search & Semantic Engine Implementation: Implemented the core SemanticSearchEngine by integrating a local Sentence Transformer model (all-MiniLM-L6-v2) to convert natural language queries into 384-dimensional dense vectors. Configured a FAISS (Facebook AI Similarity Search) IndexFlatIP vector database to perform rapid, L2-normalized nearest-neighbor retrieval, establishing a mapping system between spatial positions and primary database IDs.
	06/05/2026	10:00 PM	3:00 AM	5	API Development & Database Modeling: Created an asynchronous REST API utilizing the FastAPI framework. Designed strict

					relational database schemas using SQLAlchemy ORM and implemented data validation using Pydantic models. Developed the core POST /search endpoint, configuring it to seamlessly combine FAISS similarity rankings with dynamic SQL metadata filtering, alongside configuring CORS middleware for cross-origin access.
	08/05/2026	10:00 PM	3:30 AM	5.5	FrontEnd Development: Created a responsive design with custom CSS, focusing on fluid transitions and clear visual hierarchy. Widget Development: Built an interactive, DOM-driven JavaScript slide-out panel (widget.js) supporting complex filters, real-time query inputs, and dynamic result rendering.
	09/05/2026	3:00 PM	6:00 PM	3	Frontend-Backend REST API Integration: Connected the UI and backend layers by implementing asynchronous Javascript fetch mechanisms within the widget to communicate with the FastAPI server. Added state management to handle loading spinners, empty states, successful JSON payload parsing, and graceful error handling during network requests.
	10/05/2026	2:00 PM	4:00 PM	2	Testing and Debugging
	11/05/2026	12:00 AM	1:30 AM	1.5	Wrote the project documentation
Total no. of social hours:-				31.5	


Signature of Supervisor